Industrial Motor Maintenance Manual

Demo Document for PLANT-GPT RAG Testing

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This demonstration manual contains operating specifications, maintenance procedures, safety information, and troubleshooting data. It is intended to test document upload, indexing, retrieval, and question answering.

1. Motor Overview

The plant motor is a three-phase squirrel-cage induction motor used to drive pumps, fans, and conveyor systems. The motor should be operated within its rated electrical and mechanical limits. Routine inspection helps identify overheating, abnormal vibration, loose connections, and bearing problems before equipment failure occurs.

2. Technical Specifications

Parameter	Specification
Rated power	15 kW
Rated voltage	415 V AC, three-phase
Rated frequency	50 Hz
Rated speed	1475 RPM
Rated current	28 A
Motor efficiency	91%
Insulation class	Class F
Protection rating	IP55
Recommended ambient temperature	5°C to 40°C

3. Operating Guidelines

Before starting the motor, verify that the coupling and driven equipment are free to rotate. Confirm that the supply voltage is within the rated value and that protective devices are operational. During operation, monitor current, temperature, vibration, and unusual noise. Do not operate the motor when there is visible damage to cables, terminals, cooling fans, or the terminal box.

The recommended continuous operating temperature for the motor housing is below 80°C. If the housing temperature remains above 80°C during normal load, stop the equipment when safe to do so and investigate the cause.

4. Preventive Maintenance Schedule

Interval	Inspection / Action
Daily	Check abnormal noise, vibration, smell, and visible overheating.
Weekly	Inspect cooling openings and remove accumulated dust from external surfaces.
Monthly	Check terminal-box connections, cable condition, mounting bolts, and fan condition.
Every 500 operating hours	Inspect bearing condition and record motor current and housing temperature.
Every 2000 operating hours	Perform detailed inspection of bearings, insulation condition, coupling alignment, and mou

5. Safety Procedure

Always isolate the electrical supply before opening the terminal box or performing maintenance. Apply the plant lockout/tagout procedure and verify absence of voltage using an approved test instrument. Allow the motor to cool before touching hot surfaces. Do not bypass overload protection or operate the motor with damaged insulation or exposed conductors.

Only trained personnel should perform electrical testing, bearing replacement, alignment, or insulation-resistance measurements.

6. Troubleshooting

Symptom	Possible Cause	Recommended Check
Motor overheats	Overload, blocked cooling, high ambient temperature, bearing friction	Check motor current, cooling openings, ambient temperature, and b
Excessive vibration	Misalignment, loose mounting, worn bearing, unbalanced load	Inspect foundation bolts, coupling alignment, bearings, and driven eq
Motor does not start	No supply, protection trip, loose connection	Check incoming voltage, protective device status, and terminal con
Abnormal noise	Bearing wear, fan damage, mechanical rubbing	Inspect bearing, cooling fan, and rotating components.

7. Maintenance Records

For every maintenance activity, record the date, motor operating hours, measured current, housing temperature, vibration observation, work performed, replaced components, and technician name. Trends in temperature, current, and vibration should be reviewed during scheduled maintenance.

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